

SCAR: Constant-Size Learned Memory for Length Extrapolation

Controlled tests of recall, capacity, and memory mechanisms

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Abstract

Can a recurrent model preserve useful information beyond its training context without retaining a token-level memory? We study SCAR (State-Carrier with Attentive Recall), which combines a GRU carrier with a constant-size bank of 16 learned-decay exponential slots and an attentive read head. The corrected Study 1 release contains 135 matched runs; the cloud-only Study 2 matrix adds 172 preregistered stress-test runs spanning length extrapolation, train/test ratios, associative recall, selective copying, slot/decay mechanisms, and frozen-memory interventions. The central result is not an across-the-board accuracy win: in-distribution recall hides a separation that appears at long horizons, while multi-item retrieval and perturbation tests reveal capacity and robustness limits. The dedicated CPU benchmark falsifies the original approximately one-times-GRU cost hypothesis: SCAR is slower than a GRU but faster than RLT-lite in the measured setting. The result is a controlled synthetic study of when bounded learned memory helps, and when it does not. At the trained length, the best parity-dense result is Elman RNN ($100.0 \pm 0.0\%$), with the same rounded mean as LSTM, GRU, TokenMerge (RLT abl.), RLT, SCAR, SCAR-carrier (no memory), SCAR-no-recall. On the longest delayed-recall evaluation, SCAR reaches $100.0 \pm 0.0\%$, compared with $56.6 \pm 18.5\%$ for the carrier-only ablation and $30.5 \pm 32.4\%$ when the memory is written but not read. In the preregistered follow-up, the same model is evaluated at lengths through 4,096, reaches $72.8 \pm 13.8\%$ at the 4,096-operation endpoint, and reaches $11.4 \pm 0.6\%$ at 32 key/value pairs; selective-copy and intervention sweeps expose capacity and perturbation limits.

1 Introduction

Sequence models differ in how they carry information from the past. A recurrent network compresses history into a fixed-size state, while a causal Transformer exposes the whole prefix through an attention computation and an $O(T)$ key-value cache [1, 2, 3, 4]. Memory-augmented models make the trade-off explicit by adding a writable storage mechanism and a learned read operation [5].

This paper asks a narrower question than “which architecture is best?” A recent Recurrent Looped Transformer (RLT) design carries decoder state while also reading a growing encoder-derived memory [14]. If a small RLT-like model succeeds on state-tracking tasks, which ingredient deserves credit: the recurrent carrier, the growing memory, or the looped read path? The answer matters because these mechanisms have different inference-state costs. SCAR is a subtractive test: it keeps a gated recurrent carrier and a cheap read path, but replaces the growing memory with a fixed bank of multi-timescale exponential summaries.

Our contributions are:

1. a precisely specified constant-size state-carrier architecture with an $O(1)$ -per-token update and a fixed $O(kd)$ memory state;
2. a matched-budget, three-seed sweep that crosses architecture with supervision density and evaluates beyond the training length;
3. parameter-matched memory and read-path ablations that separate short training performance from long-delay extrapolation; and
4. a preregistered Study 2 matrix testing length ratios, associative recall, multi-item capacity, mechanisms, and interventions; and

5. a reproducible methodology audit: the earlier release had a collapsed decay initialization, a BOS train/evaluation mismatch, and an incorrect description of the Transformer baseline. All reported numbers below come from the corrected harness.

1.1 Hypotheses

We began with two falsifiable hypotheses. First, SCAR should retain the useful state-carriage behavior of RLT while avoiding its growing memory and looped decoder. Second, its fixed read path might approach the GRU’s measured cost while remaining well below RLT-lite. The results support the first hypothesis only for long-horizon extrapolation and falsify the second in its strong form: SCAR is $3.12\times$ the GRU training-step cost, although it is $0.40\times$ the RLT-lite cost.

2 Related work

Elman-style recurrent networks, LSTMs, and GRUs provide progressively more structured mechanisms for carrying state through a sequence [1, 2, 3]. Transformers replace recurrence with self-attention and expose all earlier positions during a causal forward pass [4]. External-memory networks add a separately addressable storage and read mechanism [5]; structured state-space models offer another route to long-range sequence processing with fixed-size state [6]. Segment-level recurrence and compressed memories offer related ways to extend context beyond a fixed window [11, 12]. Associative recall and its multi-query formulation provide a particularly relevant diagnostic for whether an efficient sequence model can retrieve several previously presented items [13]. Earlier work also showed that deliberately slow recurrent units can learn longer memory [7]. Linear-attention formulations expose a recurrent constant-state computation [8]; RetNet makes a related parallel/recurrent retention trade-off [9]; and Mamba uses input-dependent selective state-space updates for content-based retention [10]. SCAR is intentionally small and synthetic: it is an ablation instrument for the state-carriage versus growing-memory question, not a claim to improve language modeling or to replace these broader model families.

The RLT baseline here is a faithful small-scale implementation of the publicly described recurrent-looped pattern: a causal encoder, a recurrent decoder state, global cross-attention to encoder outputs, and a sliding-window decoder cache. It should be read as an RLT-lite baseline, not as a reproduction of every large-scale result in the original report [14].

3 SCAR

Let e_t be the embedding of token x_t and h_t the GRU carrier state:

$$h_t = \text{GRU}(e_t, h_{t-1}), \quad (1)$$

$$u_t = \sigma(W_g[e_t; h_t]) \odot h_t, \quad (2)$$

$$m_{t,i} = \lambda_i m_{t-1,i} + (1 - \lambda_i) u_t, \quad i = 1, \dots, k. \quad (3)$$

Each slot is an exponential moving average. We initialize the learned decay parameters in logit space so that λ_i spans $[0.90, 0.999]$; the corresponding half-lives span approximately 6.6 to 693 tokens. The memory therefore has a fixed $k = 16$ slots and never grows with sequence length.

The read head uses a single query from the carrier:

$$q_t = W_q \text{LN}(h_t), \quad (4)$$

$$r_t = h_t + W_o \text{softmax}\left(\frac{q_t(W_k M_t)^\top}{\sqrt{r}}\right) W_v M_t, \quad (5)$$

$$y_t = W_{\text{out}} \text{LN}(r_t), \quad (6)$$

where M_t stacks the k slots and $r = 40$ is the attention width. Each token performs a fixed amount of work independent of t and retains only the carrier and k memory slots. This gives $O(T)$ sequence work and $O(kd)$ recurrent state, in contrast to the growing $O(T)$ storage used by token-level KV memory. The constant-size memory does not make SCAR free: the dedicated timing study below measures its actual CPU cost.

4 Experimental design

4.1 Tasks and supervision

The parity task emits binary symbols and asks for their parity. The five-state task samples symbols from a fixed random transition automaton and asks for the final state. Their chance accuracies are 50% and 20%. The recall task samples the first symbol from an eight-symbol vocabulary and fills the remaining positions with independent distractors; the answer is the first symbol, giving a 12.5% chance floor. Recall is sparse-only because a repeated answer at every position would change the task into a different supervision problem.

Dense conditions supervise the state after every input symbol. Sparse conditions supervise only the final answer. All sequences are prefixed with a dedicated BOS token during both training and evaluation; sparse accuracy is measured at the final position. The main parity-sparse condition is fixed at 32 operations. A separate 4-to-8-to-16-to-32 curriculum condition was run during development, is archived under “results_curriculum/“, and is excluded from the five-condition release tables so supervision density is not conflated with curriculum.

4.2 Models and protocol

The sweep contains Elman RNN, LSTM, GRU, a full causal Transformer, TokenMerge, RLT-lite, SCAR, SCAR-carrier (memory removed), and SCAR-no-recall (memory written but never read). Widths are selected to keep models within roughly 75–92K parameters; the two SCAR ablations widen the carrier to compensate for removed modules. All models use AdamW with learning rate 3e-3, weight decay 0.01, a 200-step warmup, cosine decay, 2,500 steps, and batch size 64. The release contains 135 matched runs: nine models, five task/supervision configurations, and three seeds (0–2). All models use AdamW (learning rate 3e-3, weight decay 0.01), a 200-step warmup, cosine decay, 2,500 optimizer steps, and batch size 64. Parity and five-state runs train on 32 operations; delayed recall trains on 64. All five release conditions use fixed training lengths; the separate curriculum condition is archived outside the release tables. Evaluation uses 2,048 fresh programs per length at 16–256 operations for parity/five and 64–512 for recall. Dense supervision scores the target after every input token; sparse supervision scores one answer after the chain. Training and evaluation both prepend one BOS token and score the final position for sparse tasks.

Each evaluation point uses 2,048 fresh programs. The release reports the mean, sample standard deviation, minimum, maximum, and per-seed values for every model/configuration/length cell. The headline table is explicitly at the trained length; the second table is explicitly extrapolation.

5 Methodology audit

Before the corrected sweep, we found three errors in the experiment harness:

1. **Collapsed decay initialization.** The old code stored $\log \lambda$ but then applied a sigmoid, giving effective decays near 0.5 for every slot. The corrected code stores $\text{logit}(\lambda)$, restoring the intended multi-timescale range.
2. **Train/evaluation protocol mismatch.** Training prepended BOS but evaluation did not. The corrected evaluator applies the same prefix and scores the same final position; a protocol-sensitive regression test catches a future regression.
3. **Incorrect baseline description.** The Transformer implementation uses full causal attention, not a sliding window. The model code is unchanged; the paper, site, and tests now describe it accurately.

The earlier artifacts remain quarantined in “results_v1/“ and “data/summary_v1.json“. They are not aggregated into the release reported here.

Table 1: Accuracy (%) at the trained length (parity/five: 32 operations; recall: 64 operations), mean $\pm$ std over seeds. Chance is 50% for parity, 20% for five-state, and 12.5% for recall.

Model	Params	parity dense	parity sparse	five dense	five sparse	recall sparse
Elman RNN	76,416	100.0 $\pm$ 0.0	49.3 $\pm$ 0.5	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	13.2 $\pm$ 0.3
LSTM	75,648	100.0 $\pm$ 0.0	49.5 $\pm$ 0.6	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	13.3 $\pm$ 0.5
GRU	77,280	100.0 $\pm$ 0.0	49.5 $\pm$ 0.6	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
Transformer (full causal)	76,608	89.5 $\pm$ 7.7	49.5 $\pm$ 0.6	73.3 $\pm$ 8.6	59.8 $\pm$ 7.3	100.0 $\pm$ 0.0
TokenMerge (RLT abl.)	82,880	100.0 $\pm$ 0.0	49.5 $\pm$ 0.6	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	41.9 $\pm$ 50.3
RLT	75,000	100.0 $\pm$ 0.0	49.5 $\pm$ 0.6	100.0 $\pm$ 0.0	98.6 $\pm$ 2.4	100.0 $\pm$ 0.0
SCAR	77,896	100.0 $\pm$ 0.0	49.5 $\pm$ 0.6	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
SCAR-carrier (no memory)	77,280	100.0 $\pm$ 0.0	49.5 $\pm$ 0.6	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
SCAR-no-recall	78,710	100.0 $\pm$ 0.0	49.5 $\pm$ 0.6	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0

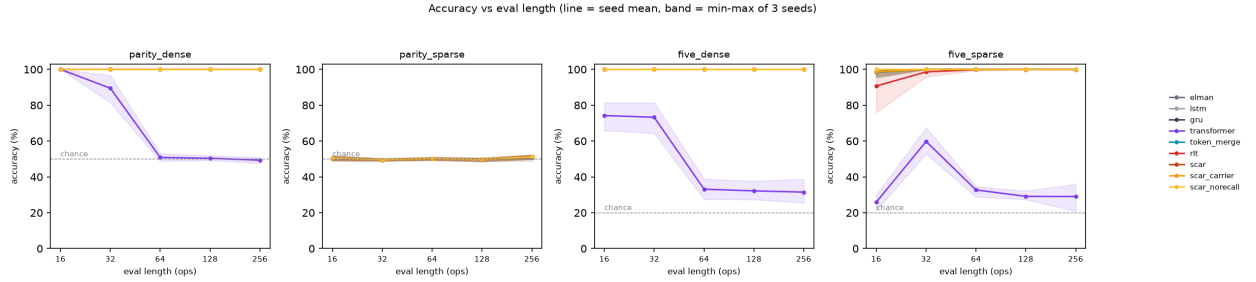


Figure 1: Accuracy versus evaluation length. Lines are seed means and shaded bands span the three-seed minimum and maximum.

6 Results

6.1 Delayed recall and ablations

At the trained length, the models that clearly beat the 12.5% chance floor are GRU, Transformer (full causal), TokenMerge (RLT abl.), RLT, SCAR, SCAR-carrier (no memory), SCAR-no-recall. The more revealing comparison is extrapolation: at 512 operations, SCAR remains at $100.0 \pm 0.0\%$, while the carrier-only and no-recall variants fall to $56.6 \pm 18.5\%$ and $30.5 \pm 32.4\%$, respectively.

The two ablations are parameter-matched to SCAR within roughly one percent: SCAR-carrier widens the GRU carrier to $d=112$ after removing the memory, while SCAR-no-recall widens d to 98 while retaining the write path. Both reach $100.0 \pm 0.0\%$ at the trained recall length, so neither component is required to fit the short training distribution. At 512 operations, however, the full model reaches $100.0 \pm 0.0\%$ versus $56.6 \pm 18.5\%$ without memory and $30.5 \pm 32.4\%$ without the read path. This supports a narrow conclusion: constant-size memory is useful for long-delay extrapolation in this probe, but the experiment does not show that the memory improves every task or every length.

7 Study 2: stress tests and failure modes

The preregistered follow-up separates the original recall headline from stress tests of length, training context, retrieval type, capacity, and memory intervention. At 4,096 operations, the recall-length sweep gives the following descriptive endpoint comparison; the model was trained at only 64 operations. The train/test ratio probes show SCAR at $100.0 \pm 0.0\%$ when trained at 32 operations and $12.2 \pm 0.8\%$ when trained at 128 operations, so the 512-token result is not treated as a universal context-length law. Associative recall is a useful negative control: SCAR reaches $11.4 \pm 0.6\%$ at 32 key/value pairs against a 6.25% single-choice chance floor, indicating that the fixed exponential summaries do not solve arbitrary multi-item retrieval.

Selective copy makes the retrieval target multi-output rather than a single first-token label. At 32 marked

Table 2: Extrapolation accuracy (%) at the longest evaluated length (parity/five: 256 operations; recall: 512 operations). No model was trained at these lengths.

Model	Params	parity dense	parity sparse	five dense	five sparse	recall sparse
Elman RNN	76,416	100.0 $\pm$ 0.0	50.7 $\pm$ 1.8	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	12.3 $\pm$ 1.9
LSTM	75,648	100.0 $\pm$ 0.0	51.1 $\pm$ 1.2	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	12.6 $\pm$ 1.7
GRU	77,280	100.0 $\pm$ 0.0	51.1 $\pm$ 1.2	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	56.6 $\pm$ 18.5
Transformer (full causal)	76,608	49.3 $\pm$ 1.6	51.1 $\pm$ 1.2	31.5 $\pm$ 6.7	29.0 $\pm$ 7.6	77.5 $\pm$ 19.7
TokenMerge (RLT abl.)	82,880	100.0 $\pm$ 0.0	51.1 $\pm$ 1.2	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	8.1 $\pm$ 7.0
RLT	75,000	100.0 $\pm$ 0.1	51.1 $\pm$ 1.2	100.0 $\pm$ 0.0	99.9 $\pm$ 0.2	100.0 $\pm$ 0.0
SCAR	77,896	100.0 $\pm$ 0.0	51.1 $\pm$ 1.2	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
SCAR-carrier (no memory)	77,280	100.0 $\pm$ 0.0	51.1 $\pm$ 1.2	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	56.6 $\pm$ 18.5
SCAR-no-recall	78,710	100.0 $\pm$ 0.0	51.1 $\pm$ 1.2	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	30.5 $\pm$ 32.4

Table 3: Study 2A delayed-recall accuracy at 4,096 operations; models were trained at 64 operations. Mean $\pm$ std over five seeds.

Model	Accuracy (%)
GRU	13.4 $\pm$ 0.6%
LSTM	12.7 $\pm$ 0.3%
Transformer (full causal)	23.3 $\pm$ 21.9%
RLT	97.4 $\pm$ 5.7%
SCAR	72.8 $\pm$ 13.8%
SCAR-carrier (no memory)	13.4 $\pm$ 0.6%
SCAR-no-recall	12.5 $\pm$ 0.4%

items, SCAR’s free-running exact-sequence accuracy is $0.0 \pm 0.0\%$ under the high-entropy distractor condition and $0.0 \pm 0.0\%$ under the low-entropy condition. The slot sweep’s 512-operation SCAR endpoints are 1: 91.1%, 2: 100.0%, 4: 77.1%, 8: 86.7%, 16: 100.0%, 32: 100.0%, 64: 89.1%. Frozen-memory interventions at 512 operations are equalize 100.0%, fastest 100.0%, noise 60.4%, shuffle 100.0%, slowest 100.0%. The learned multi-timescale decay sweep ended with final slot half-lives spanning 8.1–470.1 operations across the 16 slots, with mean read-attention entropy 2.70. These are exploratory, descriptive comparisons; no significance claim is made from the small seed counts.

8 Compute cost

On this CPU, full SCAR costs $3.12\times$ a GRU per training step, while the carrier-only ablation costs $1.04\times$. SCAR is $0.40\times$ the measured RLT training-step cost. This falsifies the original $1\times$ -GRU cost target while supporting a substantial reduction relative to RLT. These are controlled single-machine timings, not hardware-independent complexity claims.

9 Discussion

The clean fixed-length density comparison is descriptive: Under sparse parity supervision, 9 of nine architectures remain at the 50% chance floor (Elman RNN, LSTM, GRU, Transformer (full causal), TokenMerge (RLT abl.), RLT, SCAR, SCAR-carrier (no memory), SCAR-no-recall). Dense supervision makes almost every architecture look successful, whereas sparse parity leaves several at chance. With only three seeds, we report means and sample standard deviations but do not call near-equal cells significance-labeled. The strongest result is therefore not that SCAR universally wins: it is that the parameter-matched carrier-only model can match SCAR on short tasks while losing its long-delay advantage.

The combined evidence supports a conditional claim. At trained lengths, the parameter-matched SCAR ablations often fit as well as the full model; at 4,096 operations, full SCAR retains a large advantage over those

Table 4: Dedicated serial cost benchmark: 100 timed iterations after 20 warmup, torch threads = 2, batch 64, train length 32+BOS, inference length 128+BOS. Median [min-max] ms.

Model	train ms/step	×GRU	infer ms/batch
Elman RNN	23.3 [22.3–27.5]	0.75×	29.3
LSTM	29.9 [28.7–47.2]	0.96×	36.3
GRU	31.0 [30.0–46.8]	1.00×	33.2
Transformer (full causal)	32.6 [31.6–35.2]	1.05×	102.5
TokenMerge (RLT abl.)	217.6 [206.5–249.9]	7.03×	257.1
RLT	243.7 [236.2–280.8]	7.87×	391.9
SCAR	96.5 [92.6–127.0]	3.12×	104.8
SCAR-carrier (no memory)	32.2 [31.2–41.1]	1.04×	33.7
SCAR-no-recall	43.4 [42.4–63.4]	1.40×	64.7

ablations, although RLT remains stronger in that endpoint comparison. Study 2 also marks the boundary of the benefit rather than assuming that one-token recall implies general-purpose memory: associative retrieval is only modestly above chance, free-running selective copy collapses at 32 items, and the mechanism and frozen-state interventions are descriptive rather than a causal decomposition. Dense versus sparse supervision remains a methodological confounder when the protocol is not held fixed, so we do not present a universal architecture ranking or a significance claim from three seeds.

10 Limitations

This is a synthetic study with small models, a single parameter scale, three seeds for most Study 2 families, and short training budgets. The tasks test exact state tracking rather than natural-language modeling, multimodal reasoning, or noisy real-world sequences. Associative and selective-copy tasks probe more than first-token recall but do not establish language-model quality. The ablations are parameter-matched, but changing width to replace removed modules can itself change optimization and representation capacity. The RLT comparison is a small implementation, not a claim about the behavior of all RLT variants. Timing uses one CPU machine, two PyTorch threads, a parity-sized vocabulary, and an unoptimized research implementation; measured ratios should not be generalized to GPU kernels or production runtimes. Finally, the fixed memory’s success on this recall generator does not establish that learned exponential summaries are sufficient for arbitrary long-context tasks.

11 Reproducibility and release

The repository contains the model and task code (“bench.py”), regression tests (“tests/”), the quarantined Study 1 artifacts, all collected Study 2 JSON files and manifests, aggregation and chart scripts, the generated tables, and the figures used in this PDF. All optimizer steps and training timings in the two studies run on Kaggle CPU kernels through “kaggle/”; no local training is required. The release command sequence is:

```
python3 -m pytest tests/ -q
python3 aggregate.py
python3 make_charts.py
python3 -m study2.analyze study2_results --out analysis/study2
python3 study2/state_memory.py --out analysis/study2/state_memory.json
python3 make_paper.py
(cd paper && tectonic paper.tex)
```

The generated summary includes per-seed values and the curriculum flag so the reported condition cannot silently drift from the trained artifacts.

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Delayed first-token recall: accuracy vs eval length

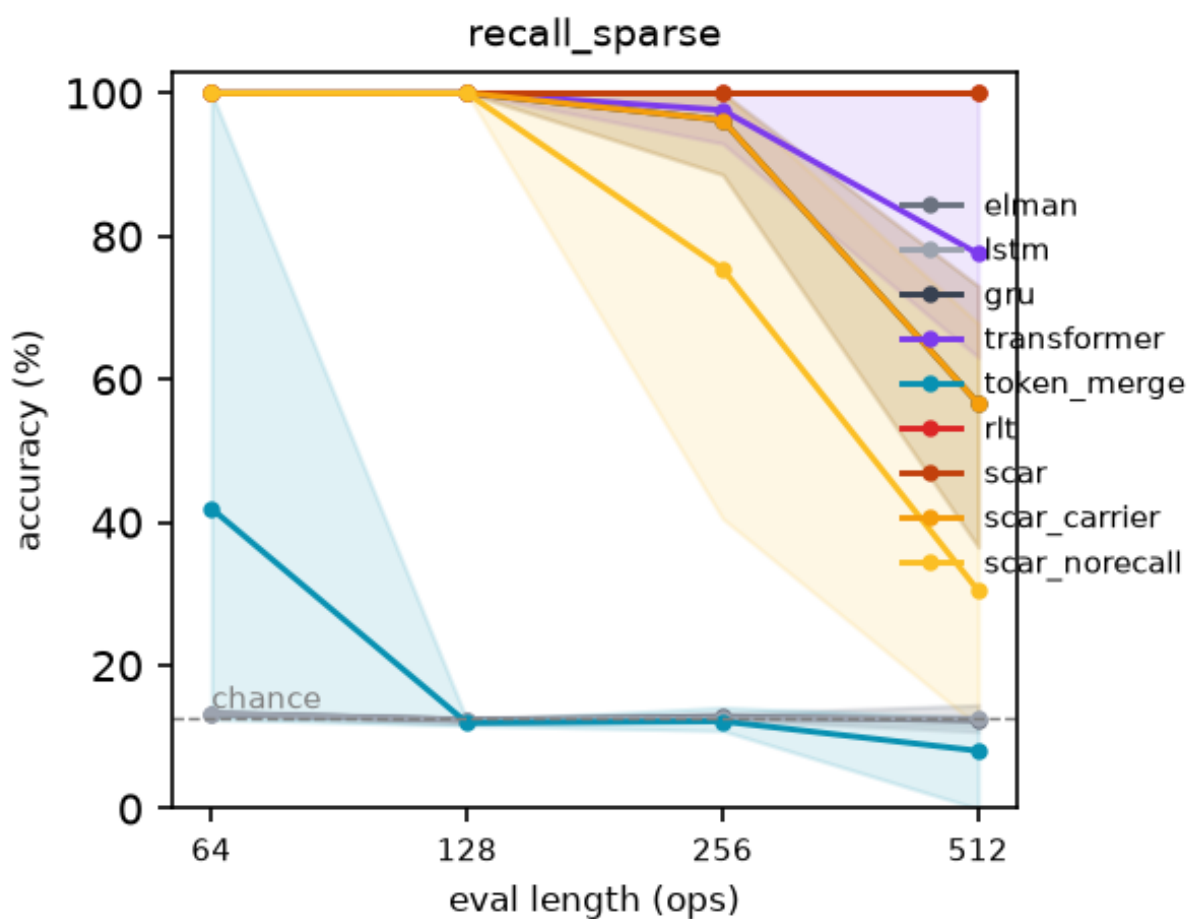


Figure 2: Delayed first-token recall across evaluation lengths; chance is 12.5%.

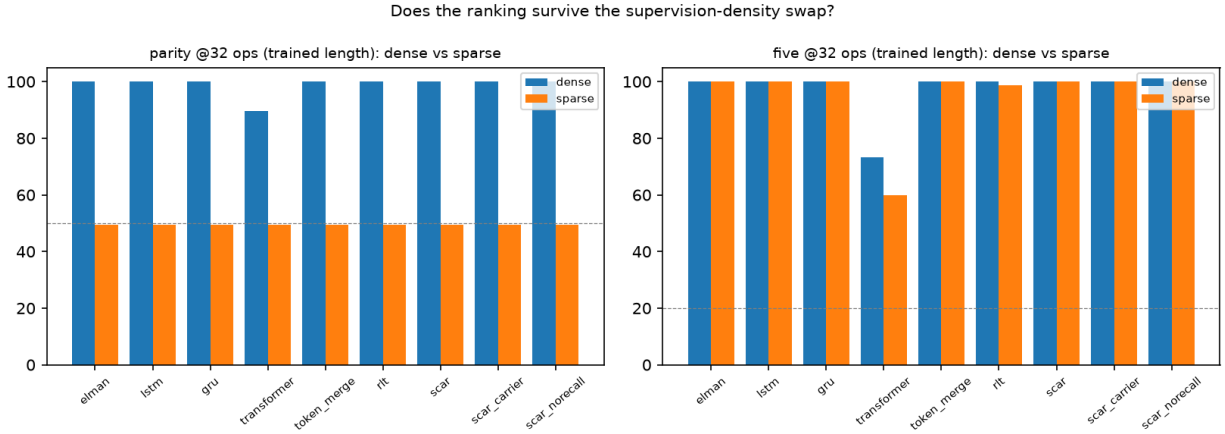


Figure 3: Changing supervision density changes which architectures escape chance on the trained-length parity and five-state tasks.

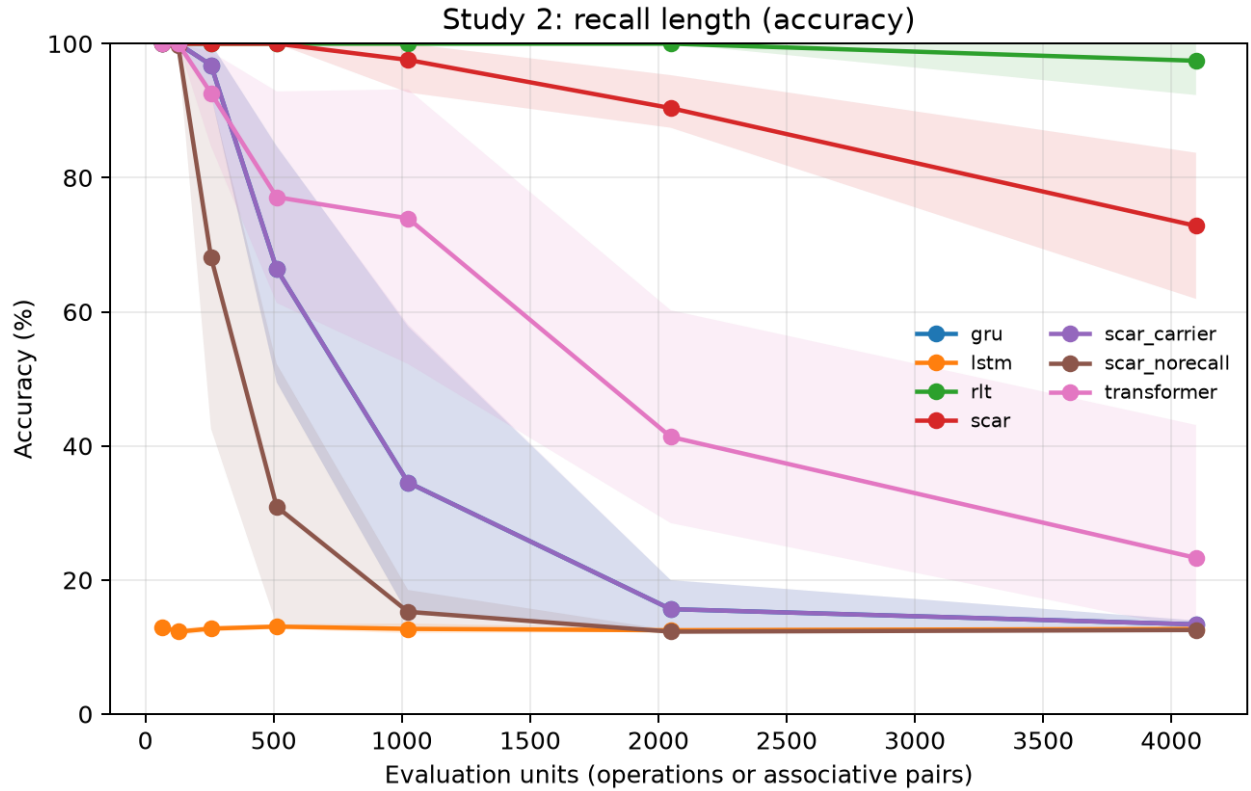


Figure 4: Study 2A: delayed-recall accuracy through 4,096 operations after training at 64 operations. Shaded bands are bootstrap intervals over seed means.

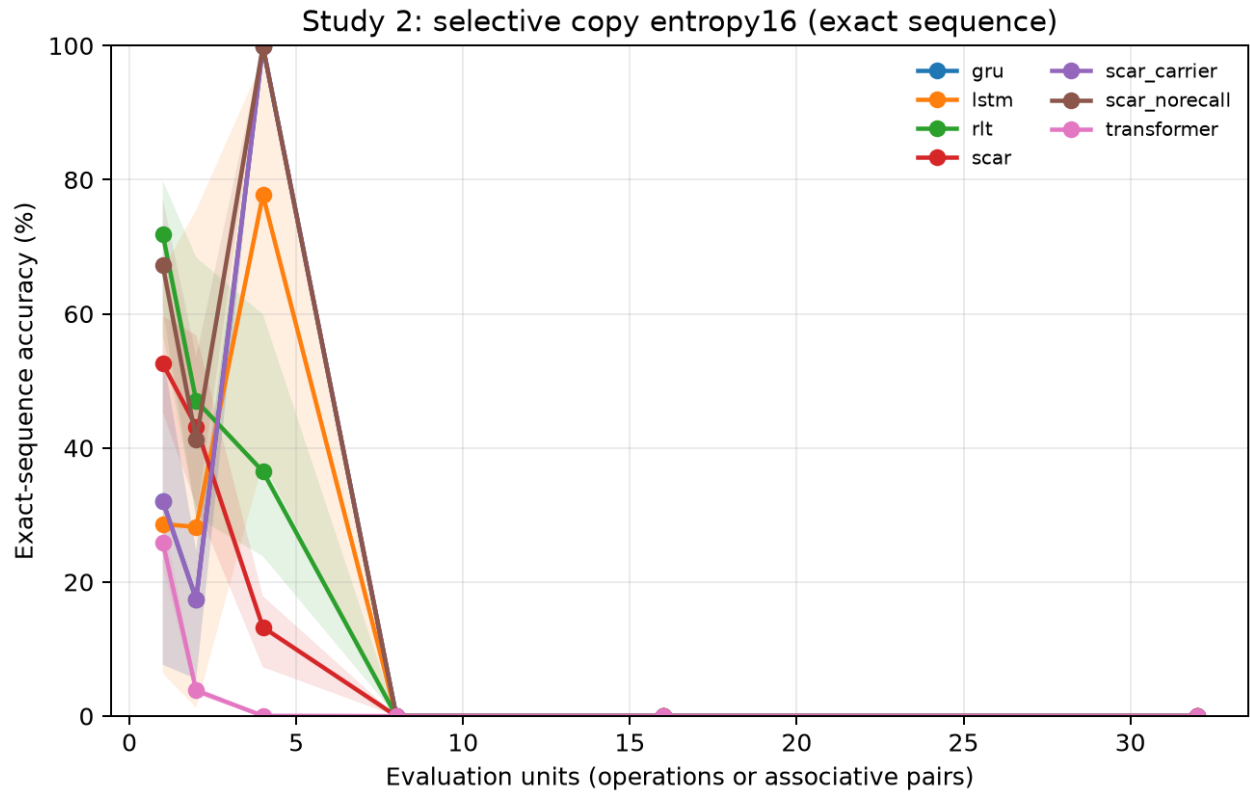


Figure 5: Study 2D: free-running exact-sequence accuracy for high-entropy selective copy.

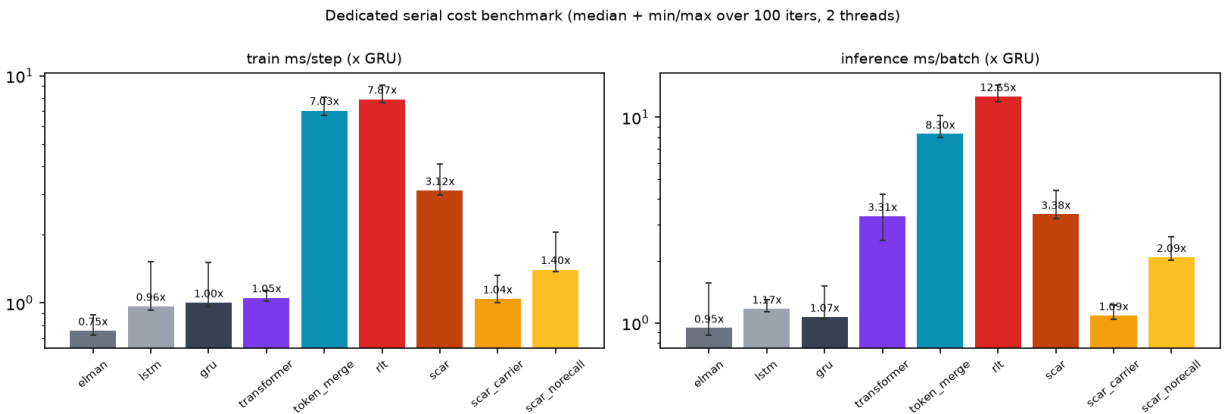


Figure 6: Dedicated serial CPU benchmark. Bars show medians and whiskers show the repeated-iteration minimum and maximum.